3.2.16

EE24BTECH11065 - spoorthi yellamanchali

Question:
The construction of triangle ABC, given that $BC = 6cm$, $\angle C = 60°$ is possible when difference of AB and AC is equal to

1) $3.2cm$
2) $3.1cm$
3) $3cm$
4) $2.8cm$

Solution:

Variable	Description
B	vertex on side of given length of the triangle
C	vertex on side of given length of the triangle
A	third vertex of the triangle

TABLE 4: Variables Used

1) $BC = 6cm$
2) $\angle C = 60°$
3) let $d = |AB - AC|$

The Law of Cosines is given by:

$$BC^2 = AB^2 + AC^2 - 2 \cdot AB \cdot AC \cdot \cos(\angle C)$$

Given $BC = 6\,\text{cm}$ and $\angle C = 60°$, we know:

$$\cos(60°) = 0.5$$

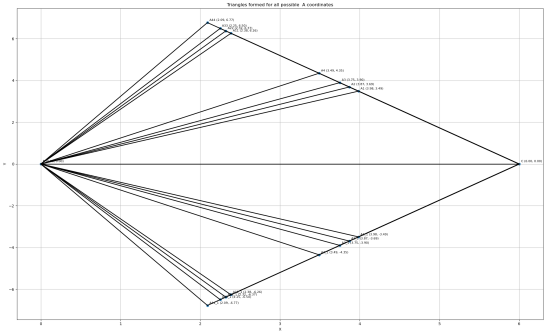
Let the difference between AB and AC be d, and set $AB = AC + d$.

Substitute these values into the Law of Cosines equation:

$$6^2 = (AC + d)^2 + AC^2 - 2(AC + d)(AC)(0.5)$$

$$AC = \frac{36 - d^2}{2d}$$

For AC to be positive, $d < 6cm$. all the options satisfy the following condition. therefore,d can take the values $3.2cm$,$3.1cm$,$3cm$,$2.8cm$. for each of the d value there are different coordinates of A.


Triangles formed for all possible A coordinates
A4 (-2.00, 6.73)
A5 (-2.00, 6.73)
A6 (-2.00, 6.73)
A7 (-2.00, 6.73)
A1 (-2.49, 4.25)
A2 (-2.49, 4.25)
A3 (-2.49, 4.25)
A8 (-2.00, 1.00)
A9 (-2.00, 1.00)
A10 (-2.00, 1.00)
(0.00, 0.00)
A11 (-2.00, -0.00)
A12 (-2.00, -0.00)
A13 (-2.00, -0.00)
A14 (-2.00, -2.25)
A15 (-2.00, -6.73)
A16 (-2.00, -6.73)
A17 (-2.00, -6.73)
A18 (-2.00, -6.73)
A19 (-2.00, -6.73)
A20 (-2.00, -6.73)
y
x